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ON

Evaluating the Effectiveness of Cybersecurity Awareness Training in Organisations

submitted in partial fulfilment of the requirements for the award of degree of

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In

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Evaluating the Effectiveness of Cybersecurity Awareness Training in Organisations**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Ajay Kumar**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

Human error continues to be one of the most significant causes of cybersecurity attacks in modern organisations. Despite the deployment of advanced security technologies such as firewalls, intrusion detection systems, and endpoint security solutions, employees remain vulnerable to phishing attacks, weak password practices, social engineering, and non-compliance with security policies.

This project evaluates the effectiveness of Cybersecurity Awareness Training (CAT) programmes in improving employee security behaviour and reducing organisational cyber risk. The study presents a mixed-methods analysis of a structured cybersecurity awareness training programme conducted over six months in a medium-sized organisation.

The project evaluates behavioural and attitudinal changes using multiple assessment techniques, including phishing simulations, knowledge assessment surveys, compliance audits, and incident reporting analysis. Various training delivery methods such as e-learning modules, simulated phishing exercises, classroom workshops, video tutorials, and gamified security challenges were compared.

Experimental results demonstrate significant improvement in employee cybersecurity awareness after training. Phishing detection rates improved by 35 percentage points, while human-attributable security incidents decreased by approximately 67%. Among all training approaches, simulated phishing exercises and gamified training modules achieved the highest effectiveness.

The findings demonstrate that structured, continuous, and interactive cybersecurity awareness training programmes can significantly reduce human-related cybersecurity risks within organisations.

Keywords: Cybersecurity Awareness Training, Phishing Detection, Security Behaviour, Social Engineering, Organisational Security, Security Compliance, Simulated Phishing.

1. Introduction

1.1 Background

Cybersecurity has become one of the most critical concerns for modern organisations due to the rapid growth of digital transformation and online communication systems. Cybercrime is expected to cost the world approximately 10.5 trillion USD annually by 2025.

Although organisations invest heavily in technical security infrastructure such as firewalls, intrusion detection systems, endpoint protection, and secure authentication systems, human behaviour continues to remain a major vulnerability.

Employees are frequently targeted through phishing attacks, social engineering techniques, poor password practices, and accidental misuse of organisational data. Studies have shown that a significant percentage of data breaches involve human error or lack of cybersecurity awareness.

To address this issue, organisations increasingly deploy Cybersecurity Awareness Training (CAT) programmes designed to educate employees about cybersecurity threats and safe online practices.

1.2 Problem Statement

Many organisations conduct cybersecurity awareness training programmes as part of compliance requirements without properly evaluating their effectiveness.

Although training programmes are widely implemented, there is limited empirical evidence regarding:

- Long-term behavioural improvement
- Reduction in phishing susceptibility
- Improvement in policy compliance
- Reduction in security incidents
- Comparative effectiveness of different training methods

Therefore, there is a need for a structured evaluation framework capable of measuring the real-world impact of cybersecurity awareness training on employee behaviour and organisational security.

1.3 Objectives

The primary objectives of this project are:

- To evaluate the effectiveness of cybersecurity awareness training programmes.
- To analyse improvements in employee phishing detection and security awareness.
- To compare multiple cybersecurity training delivery methods.
- To measure changes in security compliance and security incidents.
- To identify the most effective training approaches for organisational cybersecurity.
- To provide recommendations for designing effective cybersecurity awareness programmes.

1.4 Scope

This project focuses on evaluating cybersecurity awareness training within a medium-sized organisation consisting of approximately 350 employees.

The study evaluates multiple dimensions of cybersecurity behaviour, including phishing detection, password security, incident reporting, policy compliance, and safe browsing practices.

The project uses both quantitative and qualitative analysis methods over a six-month period.

2. Methodology

2.1 Study Design and Dataset

The study employed a mixed-methods longitudinal research design involving quantitative and qualitative analysis.

The research was conducted in a medium-sized technology services organisation with approximately 350 employees across multiple departments, including:

- Engineering
- Sales
- Operations
- Administration

The training programme was conducted over six months.

Data was collected using:

- Knowledge assessment surveys
- Simulated phishing tests
- Security compliance audits
- IT helpdesk records
- SIEM logs
- Post-training employee surveys

2.2 Data Collection Instruments

2.2.1 Knowledge Assessment Surveys

A 30-item questionnaire was used to evaluate employee understanding of:

- Phishing attacks
- Password security
- Social engineering
- Data protection
- Incident reporting procedures

2.2.2 Simulated Phishing Exercises

Monthly phishing simulations were conducted using Gophish. Metrics measured included:

- Click rate
- Credential submission rate
- Reporting rate

2.2.3 Security Incident Analysis

IT helpdesk records and SIEM logs were analysed to identify human-attributable security incidents.

2.2.4 Employee Feedback Surveys

Post-training surveys were conducted to measure:

- Training relevance
- Engagement level
- Self-efficacy
- Intent to improve security behaviour

2.3 Training Programme Design

2.3.1 E-Learning Modules

Self-paced online modules covering:

- Phishing detection
- Password hygiene
- Secure remote working
- Data protection
- Incident reporting

2.3.2 Simulated Phishing Exercises

Employees received simulated phishing emails followed by immediate corrective feedback if they clicked malicious links.

2.3.3 Classroom Workshops

Interactive workshops involving real-world cybersecurity case studies and scenario-based learning.

2.3.4 Video Tutorials

Short animated awareness videos explaining common cybersecurity threats.

2.3.5 Gamified Security Challenges

Employees participated in gamified security competitions involving quizzes, threat identification, and reporting tasks.

2.4 Statistical Analysis

- The study used:
- Paired-sample t-tests
- One-way ANOVA
- Tukey HSD tests
- Effect size analysis using Cohen's d
- The experiments were implemented in Python 3.10 using:
- SciPy
- Pandas
- Matplotlib

3. Tools and Technologies

3.1 Programming Language

Python 3.10-chosen for its extensive ML and NLP ecosystem.

3.2 Data Analysis Libraries

Pandas and NumPy

Used for:

- Data manipulation
- Statistical computation
 - Data preprocessing

SciPy

Used for:

- Hypothesis testing
- Statistical significance analysis
- ANOVA and t-tests

3.3 Cybersecurity Tools

Gophish

Used for conducting phishing simulation campaigns.

SIEM Logs

Used for analysing security incidents and employee security behaviour.

3.4 Visualisation Tools

Matplotlib

Used for generating graphs, charts, and comparative analysis.

Seaborn

Used for advanced statistical visualisations.

3.5 Development Environment

- Jupyter Notebook
- Visual Studio Code
- Git and GitHub

4. IMPLEMENTATION

4.1 System Architecture

The proposed system follows a multi-stage implementation pipeline:

- Employee Baseline Assessment
- Cybersecurity Awareness Training
- Phishing Simulation Exercises
- Behavioural Monitoring
- Statistical Analysis
- Performance Evaluation

Initially, employee cybersecurity awareness levels were measured through pre-training assessments.

Training modules were then delivered over six months using multiple delivery methods.

Behavioural data and performance metrics were continuously monitored and analysed throughout the study.

4.2 Preprocessing and Analysis

Survey responses and phishing simulation data were preprocessed and analysed using Python libraries.

The preprocessing stage included:

- Data cleaning
- Missing value handling
- Normalisation
- Statistical aggregation

Security behaviour metrics were transformed into numerical formats suitable for statistical analysis

4.3 Phishing Simulation Implementation

Monthly phishing simulation campaigns were conducted using multiple phishing email templates.

Employees who clicked phishing links received immediate feedback explaining:

- Suspicious indicators
- Social engineering tactics
- Correct security practices

Repeat clickers received additional mandatory training.

4.4 Comparative Evaluation of Training Methods

The five training approaches were compared using effectiveness scores based on:

- Employee engagement
- Knowledge retention
- Phishing resistance
- Compliance improvement
- Behavioural change

ANOVA analysis was used to determine statistically significant differences among training approaches.

4.5 Visualization and Analysis

Graphs and statistical charts were generated to visualise:

- Pre- vs. post-training performance
- Monthly phishing click reduction
- Security incident reduction
- Employee awareness distribution
- Training effectiveness comparison

5. MAJOR FINDINGS / OUTCOMES / RESULTS

5.1 Pre- vs. Post-Training Performance

The experimental analysis demonstrated statistically significant improvement across all cybersecurity behaviour dimensions.

Metric	Pre-Training	Post-Training	Improvement
Phishing Detection Rate	48%	83%	+35%
Password Strength Compliance	52%	79%	+27%
Security Policy Compliance	55%	84%	+29%
Safe Browsing Practices	61%	87%	+26%
Incident Reporting Rate	38%	71%	+33%

5.2 Phishing Click Rate Reduction

The phishing click rate steadily decreased throughout the six-month training programme.

- Baseline click rate: 52%
- Month 3 click rate: 28%
- Month 6 click rate: 17%

5.3 Training Delivery Method Effectiveness

The effectiveness scores for training methods were:

Training Method	Effectiveness Score
Simulated Phishing	89%
Gamification	86%
Classroom Workshops	81%
E-Learning Modules	74%
Video Tutorials	68%

Simulated phishing and gamification methods achieved the highest effectiveness.

5.4 Employee Awareness Distribution

Employee awareness scores improved significantly after training.

The percentage of high-risk employees scoring below 50% decreased from 38% before training to only 6% after training.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project presented a detailed evaluation of cybersecurity awareness training effectiveness within an organisational environment.

The study demonstrated that structured and continuous cybersecurity awareness programmes significantly improve employee security behaviour and reduce organisational cyber risk.

The experimental results showed:

- Significant improvement in phishing detection
- Increased policy compliance
- Improved password management
- Better incident reporting behaviour
- Reduction in human-attributable security incidents

Among all delivery methods, simulated phishing exercises and gamified security challenges proved to be the most effective.

The findings highlight the importance of interactive and behaviour-focused training approaches in strengthening organisational cybersecurity

6.2 Future Scope

Future improvements can include:

- AI-based adaptive cybersecurity training systems
- Real-time behavioural monitoring systems
- Cross-organisational comparative studies
- Long-term behavioural retention analysis
- Integration of machine learning-based risk scoring systems
- Personalised cybersecurity training modules

Future research can further improve organisational resilience against evolving cyber threats.

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APPENDICES

Appendix A: Tools Used

- Python
- Pandas
- NumPy
- SciPy
- Matplotlib
- Seaborn
- Gophish

Appendix B: Sample Workflow

- Conduct Baseline Assessment
- Deliver Cybersecurity Training
- Run Phishing Simulations
- Monitor Behavioural Metrics
- Perform Statistical Analysis
- Evaluate Training Effectiveness

Appendix C: Expected Outputs

- Improved phishing detection rates
- Reduced security incidents
- Better policy compliance
- Enhanced employee awareness
- Comparative training effectiveness reports